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RAG vs CAG

Retrieval-Augmented Generation (RAG)

What is it?

RAG is an architecture that combines information retrieval with text generation. It was proposed by Facebook AI Research (FAIR) in 2020.

How it works:

<u>Retriever:</u> Retrieves relevant documents from an external knowledge base (e.g., using FAISS on a corpus).

<u>Generator:</u> Uses a generative model (such as BART or T5) to produce an answer conditioned on both the question and the retrieved documents.

Advantages:

- Can answer questions with up-to-date data without retraining the model.
- Interpretable: you can see which documents were used.
- Scalable for QA (question answering) tasks and chatbots.

Compressed Answer Generation (CAG)

What is it?

CAG is not as standardized a term as RAG, but it may refer to:

- Compressed Answer Generation: a technique where a short and precise answer is generated using text compression strategies.

- In some papers, CAG appears as an alternative approach to RAG, where no external document retrieval occurs. Instead, the model learns to generate answers directly from the input, possibly using compressed embeddings or intermediate summaries.

Comparison

Which one to use?

- Use RAG if you need contextualized answers based on an updatable knowledge base.
- Use CAG if your task requires short, fast answers without the need for external context.

Simplified Comparison

Feature	**RAG**	**CAG**
Uses external retrieval	Yes	No
Language model	Generative (BART, T5, etc.)	Generative
Long answers	Yes (can generate them)	Generally shorter
Updatable data	Yes (easy with external bases)	Not so easy
Accuracy and context	High (thanks to documents)	Limited by input

Cuadro 1: Simplified comparison between RAG and CAG
